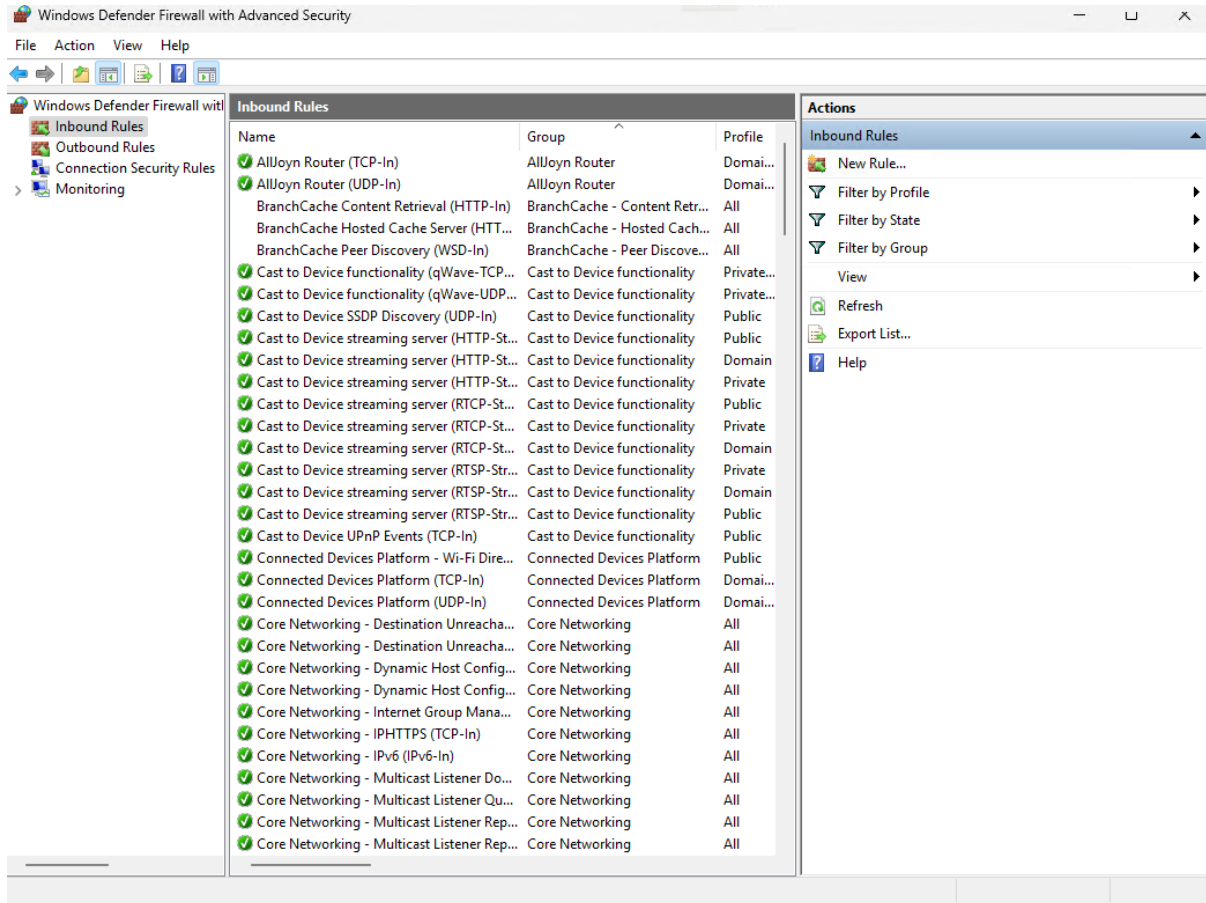
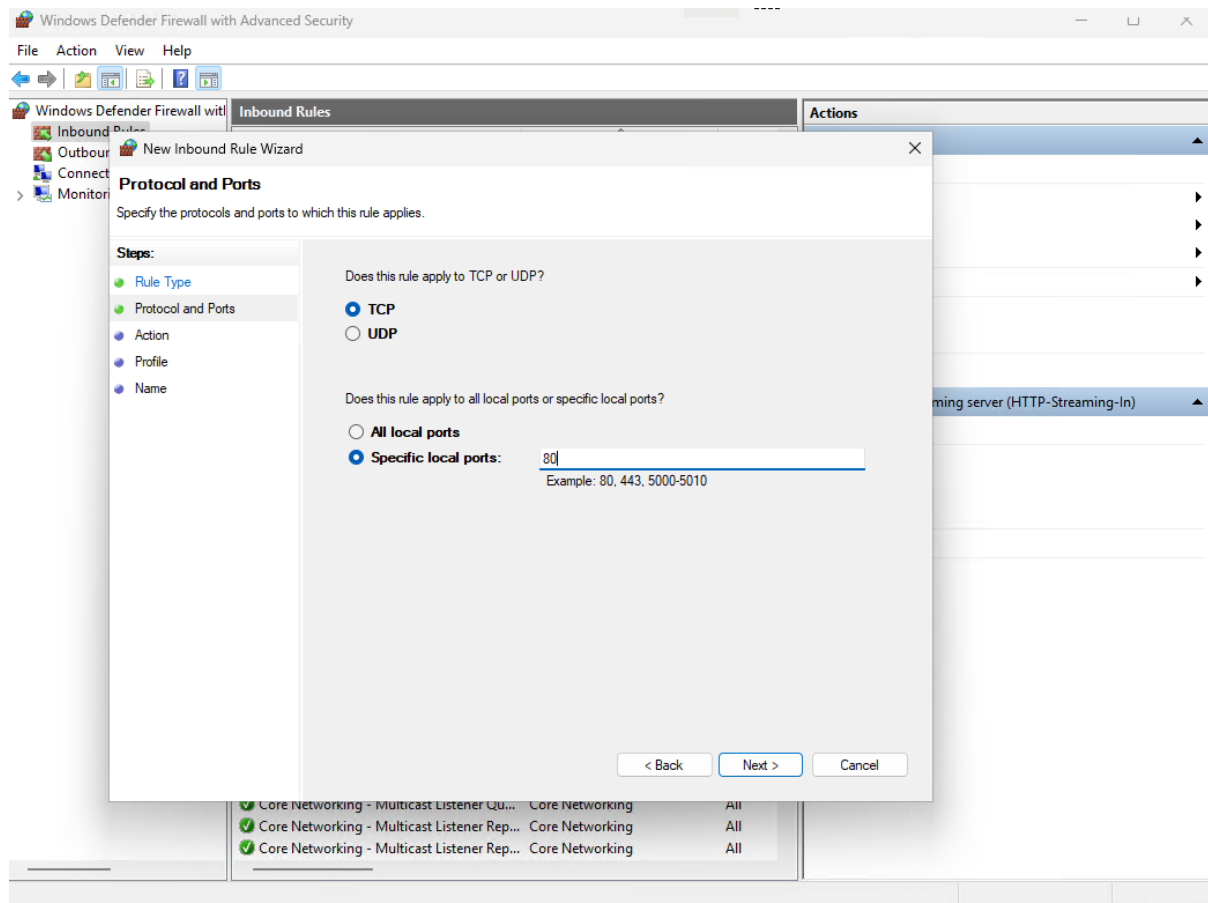


Topic: Network Firewall Configuration (Windows/Linux) – Stateful/Stateless Firewalls, Logging, and Verification..

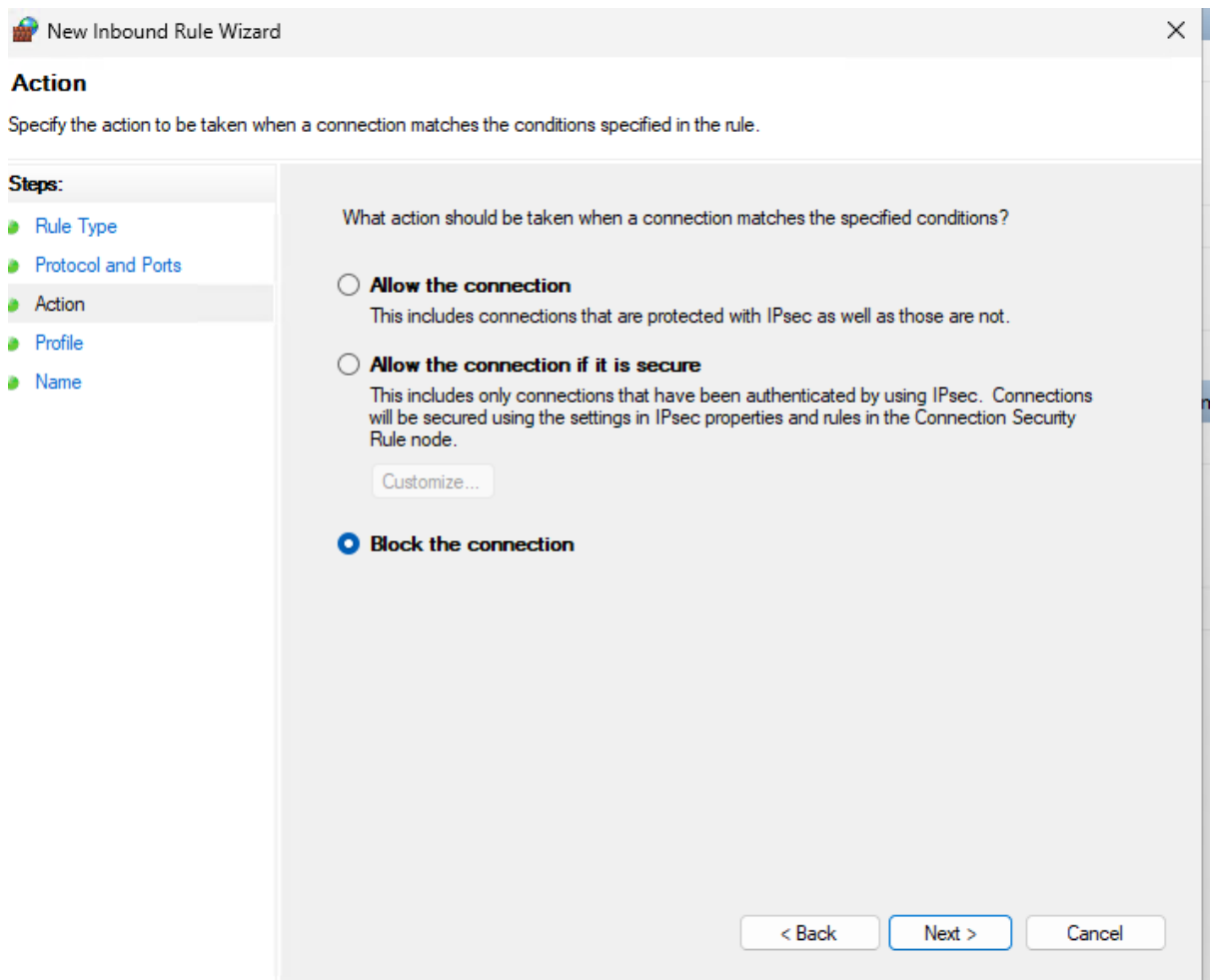
old-times-sake



- Action: Opening the "Windows Defender Firewall with Advanced Security" (WFAS) console.
- Technical Objective: To enable a quick comparison of network profiles, review existing rules, and verify the default policy before implementing changes.
- View Analysis: The screenshot displays the Inbound Rules list, showing active and inactive system rules along with their assignments to specific profiles (Domain, Private, Public) and groups.
- Success Criterion: The WFAS console was successfully launched, and all nodes (Inbound/Outbound Rules) are accessible to the administrator.



- Activity: Defining the protocol and ports within the New Inbound Rule Wizard.
- Technical Objective: Implementing a block on port TCP/80 (HTTP) to demonstrate default policy fundamentals and the process of selective service filtering (allowing vs. blocking).
- View Analysis: In the "Protocol and Ports" window, the TCP protocol was selected and the "Specific local ports" field was set to 80.
- Success Criterion: Parameters were entered according to the laboratory specifications; clicking "Next" enables the definition of the "Block the connection" action.



-
- Activity: Selecting the Action for the newly created rule within the wizard.
- Technical Objective: Implementing security policy requirements based on the explicit block principle for specific traffic flows.
- View Analysis: The "Block the connection" option was selected, ensuring that packets matching the previously defined criteria (TCP/80) are dropped/rejected.
- Success Criterion: The configuration ensures effective enforcement of the block. This should result in no response to HTTP requests and the generation of Event ID 5157 (Windows Filtering Platform packet drop) entries in the security logs during the verification phase.

New Inbound Rule Wizard

Name

Specify the name and description of this rule.

Steps:

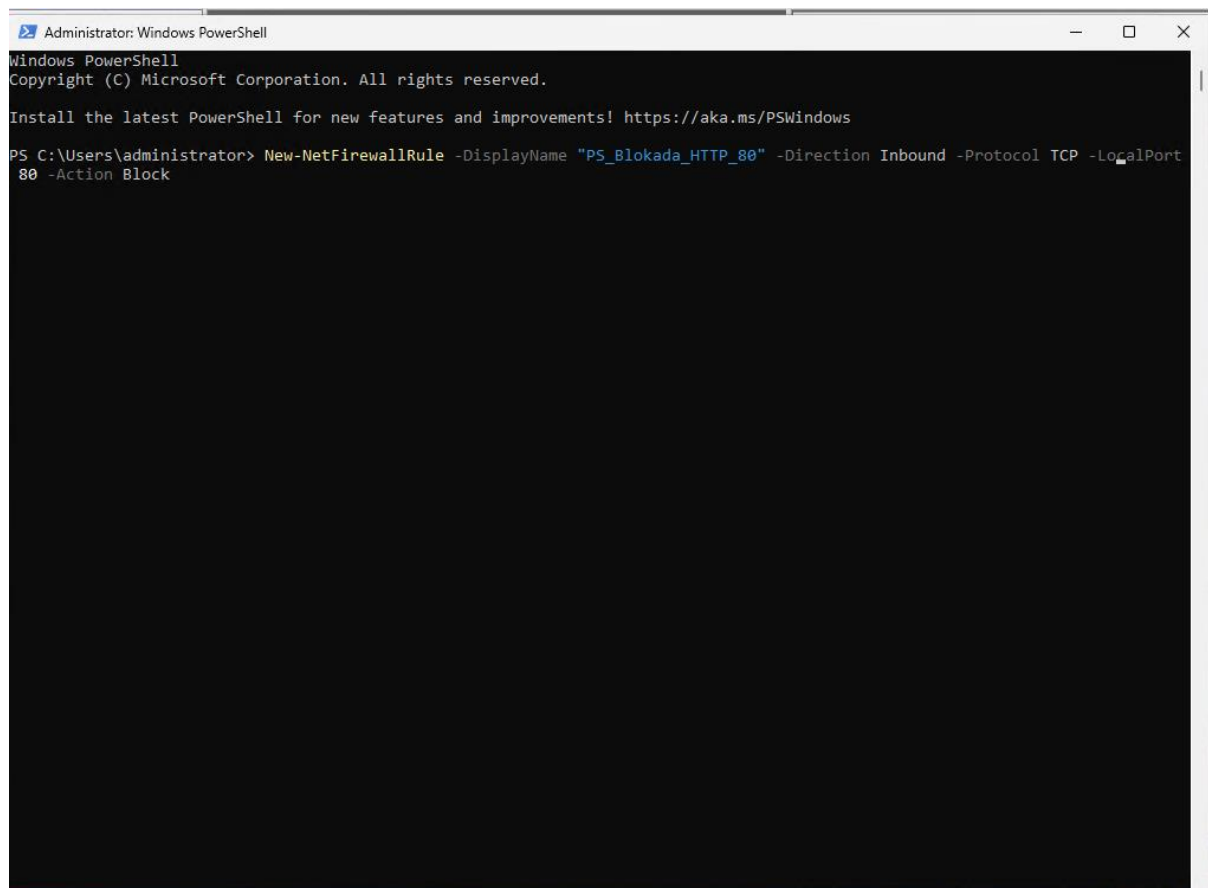
- Rule Type
- Protocol and Ports
- Action
- Profile
- Name**

Name:
Blokada_HTTP_80

Description (optional):

< Back Finish Cancel

- Activity: Assigning a unique name to the rule (Blokada_HTTP_80) and completing the creation process.
- Technical Objective: Maintaining configuration hygiene and facilitating future audits and collaborative management through clear, standardized naming conventions.
- View Analysis: A descriptive name was entered into the "Name" field. Utilizing standardized naming patterns (e.g., "Deny-HTTP") significantly improves security policy transparency for other administrators.
- Success Criterion: The rule has been successfully saved in the system and is now actively enforced by the Windows Filtering Platform (WFP).



```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\administrator> New-NetFirewallRule -DisplayName "PS_Blokada_HTTP_80" -Direction Inbound -Protocol TCP -LocalPort 80 -Action Block
```

- Activity: Utilizing the PowerShell command line to create a rule blocking the HTTP port.
- Technical Objective: Automating administrative tasks and enabling version control and documentation of configurations through scripting.
- View Analysis: The New-NetFirewallRule cmdlet is shown with parameters defining the display name, Inbound direction, TCP protocol, local port 80, and the Block action.
- Success Criterion: The command was correctly formulated and executed, allowing for rapid, repeatable deployment of identical security policies across multiple workstations or servers.

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\administrator> New-NetFirewallRule -DisplayName "PS_Blokada_HTTP_80" -Direction Inbound -Protocol TCP -LocalPort 80 -Action Block

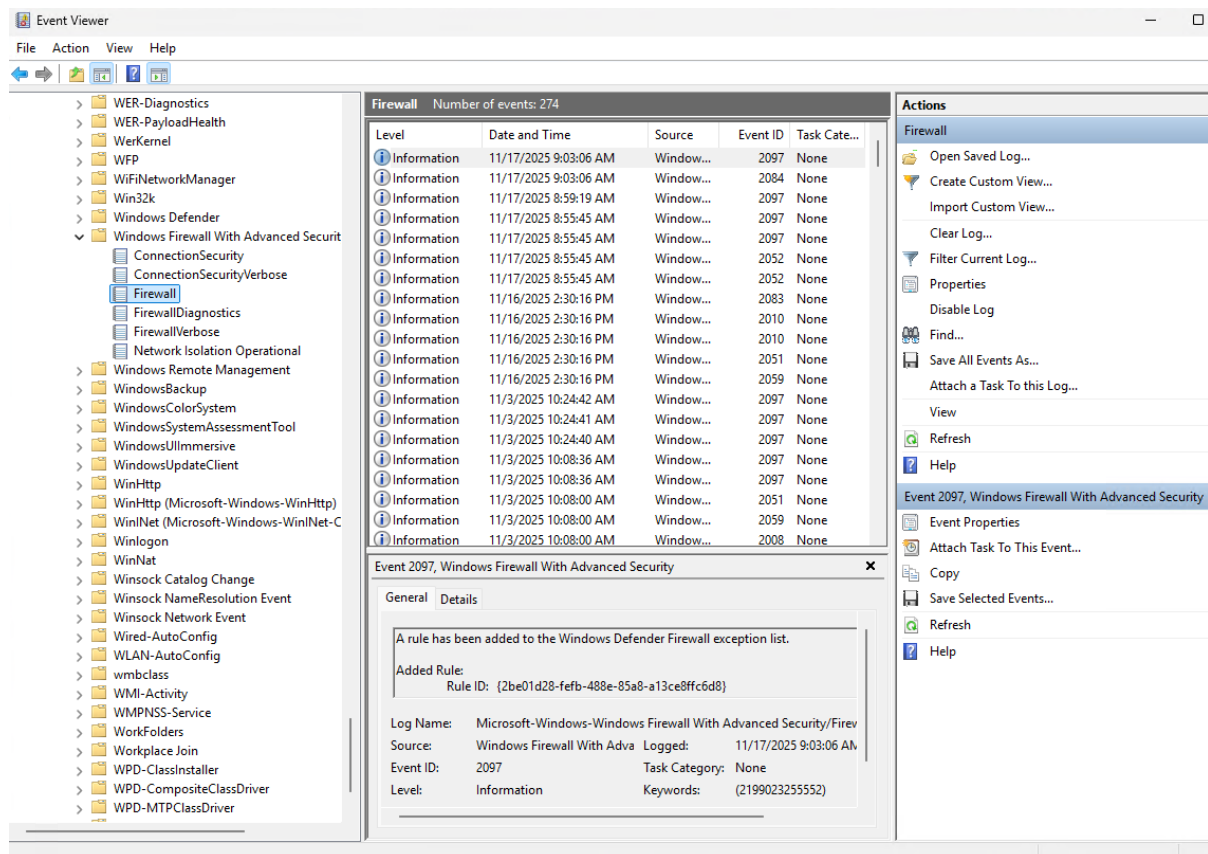
Name                               : {2be01d28-feb-488e-85a8-a13ce8ffc6d8}
DisplayName                       : PS_Blokada_HTTP_80
Description                      :
DisplayGroup                     :
Group                            :
Enabled                          : True
Profile                          : Any
Platform                        : {}
Direction                       : Inbound
Action                           : Block
EdgeTraversalPolicy              : Block
LooseSourceMapping               : False
LocalOnlyMapping                 : False
Owner                            :
PrimaryStatus                    : OK
Status                           : The rule was parsed successfully from the store. (65536)
EnforcementStatus                : NotApplicable
PolicyStoreSource                : PersistentStore
PolicyStoreSourceType            : Local
RemoteDynamicKeywordAddresses    : {}
PolicyAppId                      :
PackageFamilyName                :

PS C:\Users\administrator> _
```

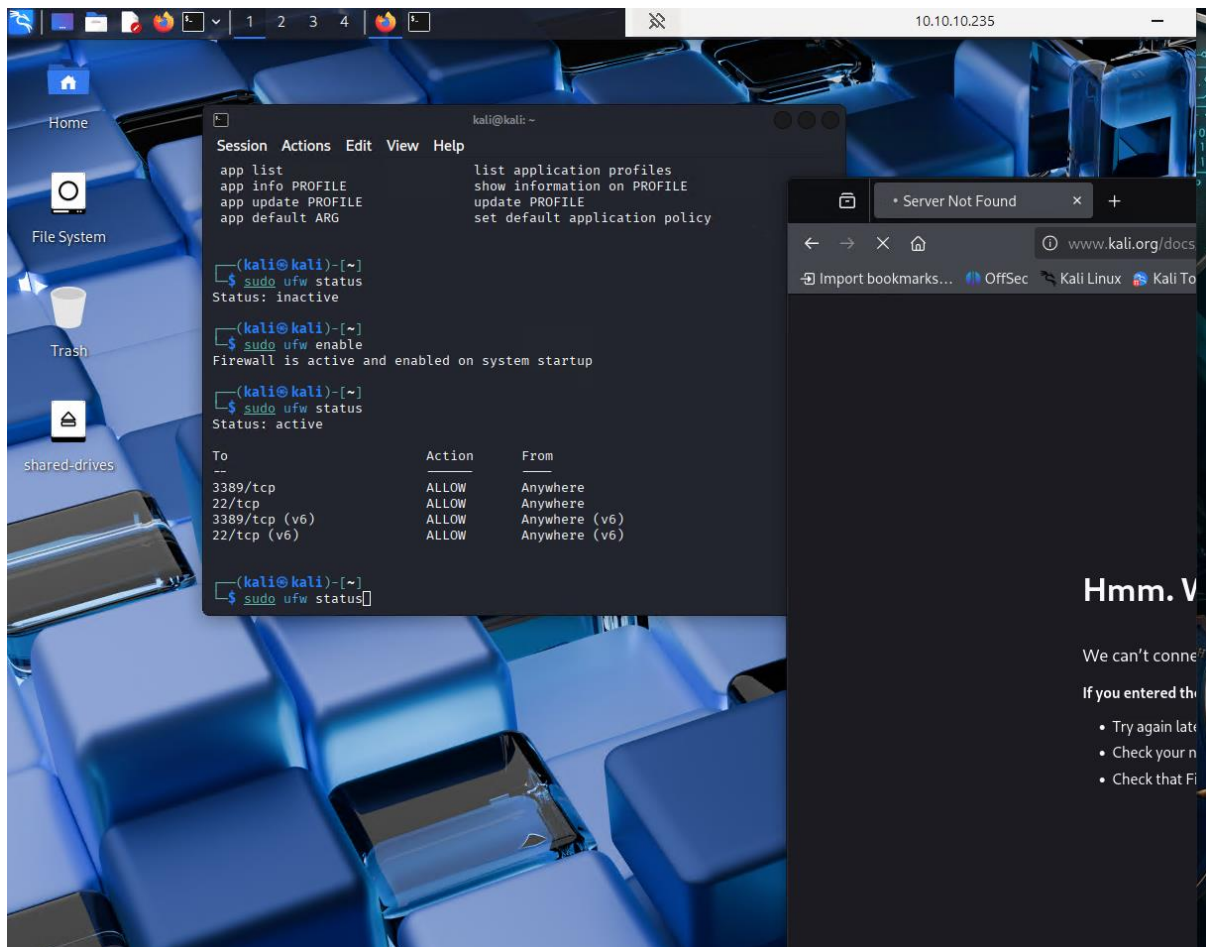
- Activity: Analyzing the script execution results within the administrator console.
- Technical Objective: Confirming that the rule was correctly parsed and stored in the system configuration store.
- View Analysis: The console displays the detailed parameters of the created rule (PS_Blokada_HTTP_80), specifically showing the statuses Enabled: True and Action: Block.
- Success Criterion: The command completed successfully (Status: OK), and the rule is now actively enforced by the system.

Inbound Rules		
Name	Group	Profile
 Blokada_HTTP_80		All
 PS_Blokada_HTTP_80		All
 AllJoyn Router (TCP-In)	AllJoyn Router	Domai...
 AllJoyn Router (UDP-In)	AllJoyn Router	Domai...
BranchCache Content Retrieval (HTTP-In)	BranchCache - Content Retr...	All
BranchCache Hosted Cache Server (HTT...	BranchCache - Hosted Cach...	All
BranchCache Peer Discovery (WSD-In)	BranchCache - Peer Discove...	All
 Cast to Device functionality (qWave-TCP...	Cast to Device functionality	Private...
 Cast to Device functionality (qWave-UDP...	Cast to Device functionality	Private...
 Cast to Device SSDP Discovery (UDP-In)	Cast to Device functionality	Public
 Cast to Device streaming server (HTTP-St...	Cast to Device functionality	Private
 Cast to Device streaming server (HTTP-St...	Cast to Device functionality	Public
 Cast to Device streaming server (HTTP-St...	Cast to Device functionality	Domain
 Cast to Device streaming server (RTCP-St...	Cast to Device functionality	Public
 Cast to Device streaming server (RTCP-St...	Cast to Device functionality	Private
 Cast to Device streaming server (RTCP-St...	Cast to Device functionality	Domain
 Cast to Device streaming server (RTSP-Str...	Cast to Device functionality	Private
 Cast to Device streaming server (RTSP-Str...	Cast to Device functionality	Domain
 Cast to Device streaming server (RTSP-Str...	Cast to Device functionality	Public
 Cast to Device UPnP Events (TCP-In)	Cast to Device functionality	Public
 Connected Devices Platform - Wi-Fi Dire...	Connected Devices Platform	Public
 Connected Devices Platform (TCP-In)	Connected Devices Platform	Domai...
 Connected Devices Platform (UDP-In)	Connected Devices Platform	Domai...
 Core Networking - Destination Unreacha...	Core Networking	All
 Core Networking - Destination Unreacha...	Core Networking	All
 Core Networking - Dynamic Host Config...	Core Networking	All
 Core Networking - Dynamic Host Config...	Core Networking	All
 Core Networking - Internet Group Mana...	Core Networking	All
 Core Networking - IPHTTPS (TCP-In)	Core Networking	All
 Core Networking - IPv6 (IPv6-In)	Core Networking	All
 Core Networking - Multicast Listener Do...	Core Networking	All
 Core Networking - Multicast Listener Qu...	Core Networking	All

- Activity: Visual verification of the active inbound rules list after completing both configuration methods (GUI and CLI).
- Technical Objective: Confirming the persistence of the implemented changes and verifying their correct precedence within the enforcement list.
- View Analysis: At the top of the list, two newly created rules are visible: Blokada_HTTP_80 and PS_Blokada_HTTP_80, both marked with the red "block" icon.
- Success Criterion: The laboratory task has been fully completed—both methods (GUI and CLI) successfully implemented the required restrictions at the firewall layer.



- Activity: Reviewing system logs to confirm firewall operations.
- Technical Objective: Ensuring accountability and observability of administrative actions by identifying changes made to the security policy.
- View Analysis: The display shows an event with Event ID 2097 from the "Firewall" channel, which confirms the addition of a new rule to the firewall exception list.
- Success Criterion: The logs provide verifiable evidence of rule enforcement, which is essential for incident response and auditing processes.



- Activity: Installing, enabling, and verifying the status of the UFW (Uncomplicated Firewall) on a Linux system.
- Technical Objective: Ensuring the persistence of changes and rule transparency while selectively permitting administrative traffic (SSH).
- View Analysis: The terminal displays the firewall transitioning from an inactive to an active state following the `sudo ufw enable` command. The final status indicates that traffic is allowed on ports 22/tcp (SSH) and 3389/tcp (RDP) from any source.
- Success Criterion: The UFW firewall correctly enforces an "explicit allow" policy for critical administrative services while simultaneously blocking all other traffic not specified in the rules.

```

Microsoft Windows [Version 10.0.26100.1742]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator>cd c:\apache24

c:\Apache24>cd \bin
The system cannot find the path specified.

c:\Apache24>cd .\bin

c:\Apache24\bin>httpd.exe -t

c:\Apache24\bin>
c:\Apache24\bin>
c:\Apache24\bin>
c:\Apache24\bin>
c:\Apache24\bin>httpd.exe -t
Syntax OK

c:\Apache24\bin>httpd.exe -k install -n "Apache HTTP Server"
Installing the 'Apache HTTP Server' service
The 'Apache HTTP Server' service is successfully installed.
Testing httpd.conf....
Errors reported here must be corrected before the service can be started.

c:\Apache24\bin>cd .\bin
The system cannot find the path specified.

c:\Apache24\bin>cd C:\
C:\>httpd.exe -k install -n "Apache HTTP Server"
'httpd.exe' is not recognized as an internal or external command,
operable program or batch file.

C:\>cd .\bin
The system cannot find the path specified.

C:\>cd c:\apache24

c:\Apache24>httpd.exe -k install -n "Apache HTTP Server"
'httpd.exe' is not recognized as an internal or external command,
operable program or batch file.

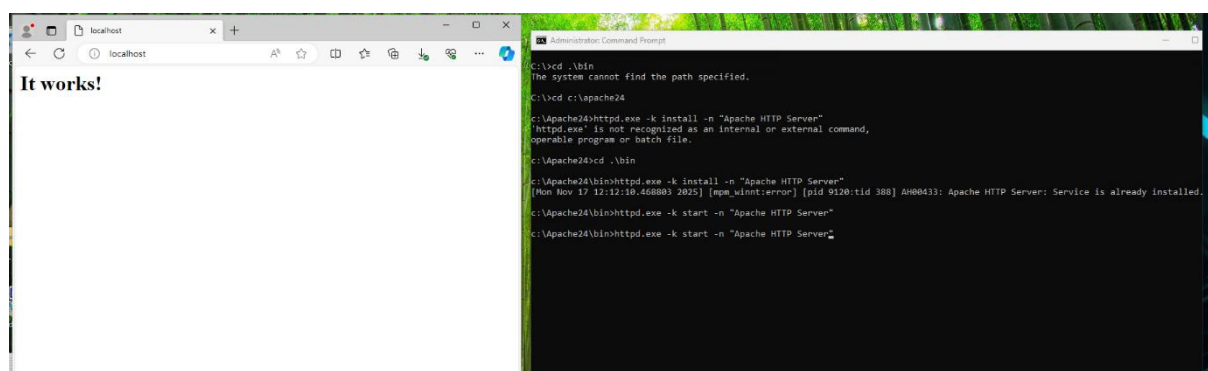
c:\Apache24>cd .\bin

c:\Apache24\bin>httpd.exe -k install -n "Apache HTTP Server"
[Mon Nov 17 12:12:10.468803 2025] [mpm_winnt:error] [pid 9120:tid 388] AH00433: Apache HTTP Server: Service is already installed.

c:\Apache24\bin>

```

- Activity: Verifying the Apache HTTP Server configuration syntax prior to service startup.
- Technical Objective: Ensuring that the web server service—the intended target for port 80 blocking tests—is functioning correctly at the application layer.
- View Analysis: Execution of the `httpd.exe -t` command within the Apache binary directory, which returned the status message: "Syntax OK."
- Success Criterion: The service is prepared for both positive and negative testing. Any subsequent lack of connectivity will be definitively attributed to firewall enforcement rather than an application-level misconfiguration.



- Activity: Performing a positive connectivity test with the HTTP server prior to activating the blocking rules.

- Technical Objective: Confirming the proper functioning of the network stack and the application under non-restrictive conditions.
- View Analysis: The web browser displays the default "It works!" landing page, which confirms that port 80 is accessible and the service is responding.
- Success Criterion: The test environment is fully operational, providing a reliable baseline for evaluating the effectiveness of firewall rules in the subsequent steps.